MATH 2241 INTRODUCTION TO MATHEMATICAL ANALYSIS, FALL SEMESTER 2025
CHAPTER 3: LIMITS AND CONTINUITY

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Goal of this section:

(1) Continuous functions
(2) Construction of functions
(3) Examples (Showing some known functions to be continuous)
(4) Continuous functions on an interval: boundedness, maximum-minimum theorem, preserving intervals
(5) Uniform Continuity

1. Continuous functions

1.1. **Definition.**

Definition 1.1. (1) Let $A \subset \mathbb{R}$, let $f : A \to \mathbb{R}$. Let $c \in A$. We say that f is continuous at c if for any given $\epsilon > 0$, there exists $\delta > 0$ such that for any $x \in A$ with $|x - c| < \delta$,

$$|f(x) - f(c)| < \epsilon.$$

(2) If f is not continous at c, we also say that f is discontinuous at c.
(3) For a subset $B \subset A$, we say that f is continuous on B if f is continuous at every point of B.

Remark 1.2. One may compare with the concept of limit. The above definition means if $c \in A$ is a cluster point, $\lim_{x\to c} f$ exists. In addition to that, we further require $\lim_{x\to c} f(x) = f(c)$. If a point c in A is not a cluster point, we also call the point to be an isolated point in A. Isolated points do not have much to be investigated in continuity (see the following exercise).

Exercise 1.3. Let $f : A \to \mathbb{R}$. Let $c \in A$ be an isolated point in A. Prove that f is continuous at c by $\epsilon - \delta$ definition.

Using Remark 1.2 and Exercise 1.3, we have:

Exercise 1.4. Let $f : A \to \mathbb{R}$. Let $c \in A$. Then f is continuous at c if and only if either

(1) c is a cluster point of A and $\lim_{x\to c} f(x) = f(c)$;
(2) c is an isolated point of A.

Exercise 1.5. (Sequential criteria) Let $f : A \to \mathbb{R}$, let $c \in A$. Then f is continuous at c if and only if for every sequence (x_n) in A that converges to c, the sequence $(f(x_n))$ converges to $f(c)$.

Exercise 1.6. Let $f : A \to \mathbb{R}$, let $c \in A$. Then f is discontinuous at c if and only if there exists a sequence (x_n) in A that converges to c, the sequence $(f(x_n))$ does not converge to $f(c)$.

1.2. **Examples.** Several typical examples can be shown to be continuous by the previous discussions on limits.

Example 1.7.
(1) (Constant function) For $b \in \mathbb{R}$, let $f(x) = b$. Then, for any $c \in \mathbb{R}$, $\lim_{x\to c} f = b = f(c)$. Hence, f is continuous at any c.
(2) Let $f(x) = x$. For any $c \in \mathbb{R}$, $\lim_{x\to c} f(x) = c = f(c)$. Hence, f is continuous at every point in $\mathbb{R}$.

Example 1.8. (Dirichlet discontinuous function) Let $f : \mathbb{R} \to \mathbb{R}$. Define

$$f(x) = \begin{cases} 1 & \text{if } x \text{ is rational} \\ 0 & \text{if } x \text{ is irrational} \end{cases}$$

Then f is discontinuous at any point in $\mathbb{R}$.
Proof: Let $c \in \mathbb{R}$. We consider two cases:

(a) Suppose c is rational. Then, for any $n \in \mathbb{N}$, let $x_n = c + \frac{\sqrt{2}}{n}$. Then x_n is irrational for all $n \in \mathbb{N}$. Hence, $f(x_n) = 0$ for all $n \in \mathbb{N}$ and so $(f(x_n))$ does not converge to $f(c) = 1$. By Exercise 1.6, f is discontinuous at c.
(b) Suppose c is irrational. Then, for any $n \in \mathbb{N}$, let (x_n) be a sequence of rational numbers with $\lim_{n\to\infty} x_n = c$. Hence, $f(x_n) = 1$ for all $n \in \mathbb{N}$ and so $(f(x_n))$ does not converge to $f(c) = 0$.

Exercise 1.9. For $x \in \mathbb{R}$, define $[[x]]$ to be the greatest integer n such that $n \leq x$. Determine the points of continuity of the following functions:

(1) $f(x) = [[x]]$
(2) $f(x) = [[\sin(x)]]$

1.3. **Extending a continuous function.** Let $f : A \to \mathbb{R}$ be continuous. Let c be a cluster point of A. Suppose $c \notin A$. If $\lim_{x\to c} f$ exists, then f can be extended to

a continuous function $\widetilde{f} : A \cup \{c\} \to \mathbb{R}$ such that:

$$\widetilde{f}(x) = \begin{cases} f(x) & x \in A \\ \lim_{x\to c} f & x = c \end{cases}$$

Example 1.10. Let

$$f(x) = x \sin\left(\frac{1}{x}\right)$$

defined for $x \neq 0$. For $x \neq 0$, since

$$-|x| \leq x \sin\left(\frac{1}{x}\right) \leq |x|,$$

the squeeze theorem gives that

$$\lim_{x\to 0} x \sin\left(\frac{1}{x}\right) = 0.$$

We can extend f to the function

$$\widetilde{f}(x) = \begin{cases} x \sin\left(\frac{1}{x}\right) & x \neq 0 \\ 0 & x = 0 \end{cases}$$

which is continuous at 0.

Exercise 1.11. Let $f(x) = \frac{x^2+x-12}{x-3}$ $(x \neq 3)$. Extend f to a functon defined also in $x = 3$.

2. Construction of continuous functions

2.1. **Basic operations.**

Theorem 2.1. *Let $c \in A$. Let $f, g : A \to \mathbb{R}$ be continuous functions at c. Let $b \in \mathbb{R}$. Then*

1. *$f + g$, $f - g$, bf and fg are continuous functions at c; and*
2. *if $g(c) \neq 0$, then $\frac{f}{g}$ is continuous at c.*

Proof. As shown before, if c is an isolated point, then there is nothing to prove. Suppose c is a cluster point.

$$\lim_{x\to c}(f \pm g) = \lim_{x\to c} f \pm \lim_{x\to c} g = f(c) \pm g(c) = (f \pm g)(c),$$
$$\lim_{x\to c}(fg) = (\lim_{x\to c} f)(\lim_{x\to c} g) = f(c)g(c) = (fg)(c).$$

Since we also have $g(c) \neq 0$,

$$\lim_{x\to c}\left(\frac{f}{g}\right) = \frac{\lim_{x\to c} f}{\lim_{x\to c} g} = \frac{f(c)}{g(c)}.$$

□

Exercise 2.2. Deduce bf is a continuous function from that fg is a continuous function.

2.2. **Composition of functions.** For a function $f : A \to \mathbb{R}$ and a function $g : B \to \mathbb{R}$, suppose $f(A) \subset B$ and define

$$(g \circ f)(x) = g(f(x)).$$

Theorem 2.3. *Let $f : A \to \mathbb{R}$ and let $g : B \to \mathbb{R}$. Suppose $f(A) \subset B$. If f is continuous at $c \in A$ and g is continuous at $f(c)$, then $g \circ f$ is continuous at c.*

Proof. Let (x_n) be a sequence converging to c. By the sequential criteria (Exercise 1.5), $(f(x_n))$ converges to $f(c)$. By the sequential criteria again, the sequence $(g(f(x_n)))$ also converges to $g(f(c))$. Since (x_n) is an arbitrary sequence converging to c, the sequential criteria also implies that $(g(f(x_n)))$ also converges to $g(f(c))$. □

Exercise 2.4. Prove Theorem 2.3 by using the ϵ-δ definition of continuity.

3. Examples of continuous functions

Example 3.1. (a) Any polynomial functions are continuous. It is obtained by scalar multiplication, addition and compositions from the continuous function $f(x) = x$ (Example 1.7).

(b) Let $p(x)$ and $q(x)$ be polynomials. Suppose $q(c) \neq 0$. Let $f(x) = \frac{p(x)}{q(x)}$. It is shown before that, if $q(c) \neq 0$,

$$\lim_{x \to c} \frac{p(x)}{q(x)} = \frac{\lim_{x \to c} p(x)}{\lim_{x \to c} q(x)} = \frac{p(c)}{q(c)}.$$

Hence, f is continuous at c.

(c) Let $f(x) = \sin(x)$. We shall use the following two algebraic properties:

$$|\sin(x)| \leq |x|,$$

$$|\cos(x)| \leq 1,$$

and

$$\sin(x) - \sin(y) = 2 \sin\left(\frac{1}{2}(x-y)\right) \cos\left(\frac{1}{2}(x+y)\right).$$

Then,

$$\begin{aligned} |\sin(x) - \sin(c)| &= 2|\sin\left(\frac{1}{2}(x-c)\right) \cos\left(\frac{1}{2}(x+c)\right)| \\ &\leq |x - c| \end{aligned}$$

Since $\lim_{x \to c} |x - c| = 0$, the squeeze theorem implies that

$$\lim_{x \to c} |\sin(x) - \sin(c)| = 0.$$

and so

$$\lim_{x \to c} \sin(x) = \sin(c).$$

Thus, sin is continuous on $\mathbb{R}$.

Remark. As an extension of Example 3.1, for $n, m \in \mathbb{N}$, $f(x) = x^{m/n}$ is also continuous functions. A rigorous proof requires more than what we discussed so far. But one may freely use this later.

Exercise 3.2. Let $f(x) = \sqrt{x + \sqrt{x}}$ for $x \geq 0$. Use Theorems 2.1 and 2.3 to explain that f is continuous on $[0, \infty)$.

Example 3.3. (1) Show that $f(x) = |x|$ is continuous.
(2) Let $f : A \to \mathbb{R}$ be continuous. Define $|f| : A \to \mathbb{R}$ given by $|f|(x) = |f(x)|$ for all $x \in A$. Show that $|f|$ is continuous.

Exercise 3.4. Let f and g be continuous functions. Let $h(x) = \sup\{f(x), g(x)\}$. Then h is a continuous function. (Hint: use $h(x) = \frac{f(x)+g(x)}{2} + \frac{|f(x)-g(x)|}{2}$.)

Exercise 3.5. (Composition of discontinuous function and a continuous function) Given an example of $f : \mathbb{R} \to \mathbb{R}$ which is discontinuous at every point and $|f|$ is continuous at every point.

4. Boundedness of continuous functions

Definition 4.1. A function $f : A \to \mathbb{R}$ is said to be bounded on A if there exists a number $M > 0$ such that $|f(x)| < M$ for all $x \in A$. If a function f is not bounded, we also usually say that f is unbounded on A.

Exercise 4.2. Let f be a unbounded function. Show that there exists a sequence (x_n) on A such that $|f(x_n)| > n$.

We now prove a boundedness for a continuous function in a closed bounded interval. The application

Theorem 4.3. *(Boundedness of a continuous function on a closed bounded interval) Let I be a closed bounded interval. Let $f : I \to \mathbb{R}$ be a continuous function. Then f is bounded on I.*

Proof. One proves by contradiction. Suppose f is unbounded.
Step 1: Obtain an unbounded sequence
By Exercise 4.2, we can find a sequence (x_n) in I such that

$$(*) \quad |f(x_n)| > n.$$

Step 2: Obtain a convergent subsequence
Since I is closed, (x_n) is bounded. Now Bolzano-Weierstrass Theorem implies that there exists a convergent subsequence (x_{n_k}) of (x_n). By Exercise 1.5, $(f(x_{n_k}))$ is also convergent, which implies that $f(x_{n_k})$ is bounded. Hence, $|f(x_{n_k})| < M$ for a

number $M > 0$.

Step 3: Obtain a contradiction
By Step 1, the subsequence $|f(x_{n_k})| > n_k$ and so $(f(x_{n_k}))$ is unbounded. This contradicts Step 2. □

Exercise 4.4. Find an example that $I = [a, \infty)$ and $f : I \to \mathbb{R}$ is a continuous function, but f is not bounded on I.

Exercise 4.5. Find an example that $I = (a, b]$ for some $a, b \in \mathbb{R}$, and $f : I \to \mathbb{R}$ is a continuous function, but f is not bounded on I.

Exercise 4.6. Find an example that I is a closed interval and $f : I \to \mathbb{R}$ is a continuous function, but f is not bounded on I.

5. The maximum-minimum theorem (a.k.a. Extreme Value Theorem)

5.1. **The maximum-minimum theorem.**

Definition 5.1. Let $f : A \to \mathbb{R}$. We say that f has an absolute maximum on A if there exists $x^* \in A$ such that for all $x \in A$,

$$f(x^*) \geq f(x).$$

We say that f has an absolute minimum on A if there exists $x_* \in A$ such that

$$f(x_*) \leq f(x).$$

Example 5.2.
(1) Let $f : [-\frac{1}{2}, 1] \to \mathbb{R}$ given by $f(x) = x^2$. Then f has the absolute maximum at $x = 1$ and f attains the absolute minimum at $x = 0$.
(2) Let $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = \sin(x)$. Then f has the absolute maximum at points $x = 2n\pi + \frac{\pi}{2}$ $(n \in \mathbb{Z})$ and the absolute minimum at points $x = 2n\pi - \frac{\pi}{2}$ $(n \in \mathbb{Z})$.

In general, a continuous function may not have an absolute maximum/minimum:

Exercise 5.3.
(1) Give an example of a continuous bounded function $f : (0, 1) \to \mathbb{R}$ which does not attain an absolute maximum.
(2) Give an example of a continuous function $f : [0, \infty) \to \mathbb{R}$ which does not attain an absolute minimum.

Theorem 5.4. *(Maximum-minimum theorem) Let I be a closed bounded interval and let $f : I \to \mathbb{R}$ be a continuous function. Then f has absolute maximum and absolute minimum.*

Proof. We only prove the statement for absolute maximum, and a proof for the statement for absolute minimum is similar.
Step 1: Use $\sup f(I)$ to construct a sequence (x_n)

By Theorem 4.3, $f(I)$ is bounded. Then $\sup f(I)$ exists and let $s^* = \sup f(I)$. For any $n > 0$, there exists $x_n \in I$ such that $s^* - \frac{1}{n} < f(x_n) \leq s^*$.

Step 2: use Bolzano-Weierstrass Thm to construct a convergent seqeunce
Now, by the Bolzano-Weierstrass Theorem, there exists a convergent subsequence (x_{n_k}) of (x_n). Let $x^* = \lim_{k\to\infty} x_{n_k}$.

Step 3: Show $x^* \in I$
Moreover, let $I = [a, b]$ and then we have $a \leq x_{n_k} \leq b$ for all $k \in \mathbb{N}$, and so it is shown in Chapter 2 that

$$a \leq x^* \leq b$$

and so $x^* \in I$.

Step 4: Show $f(x^*) = s^*$
We have the inequality $s^* - \frac{1}{n_k} < f(x_{n_k}) \leq s^*$ for all k. Since $\lim_{k\to\infty}(s^* - \frac{1}{n_k}) = s^*$, the sqqueeze theorem gives that

$$\lim_{k\to\infty} f(x_{n_k}) = s^*$$

By limit criteria of continuity (Exercise 1.5), we have $f(x^*) = s^*$. □

Exercise 5.5. Let $I = [a, b]$, and let $f : I \to \mathbb{R}$ be a continuous function such that $f(x) > 0$ for all $x \in I$. Prove that there exists $m > 0$ such that $f(x) \geq m$ for all $x \in I$.

5.2. **Bolzano's intermediate theorem.** We now move to some intermediate values between the absoluate maximum and the absolute minimum. One may also rephrase to the question to find the roots of a function f i.e. solve $f(x) = 0$. The following theorem and its proof indicate a method, called the bisection method. Application to a continous function on an interval will be given in Corollary 5.11.

Theorem 5.6. *Let $I = [a, b]$ and let $f : I \to \mathbb{R}$ be a continuous function. If either $f(a) < 0 < f(b)$ or $f(a) > 0 > f(b)$, then there exists $c \in I$ such that $f(c) = 0$.*

Proof. Step 1: Construct sequences by bisections
We only consider the case that $f(a) < 0 < f(b)$. We shall prove by contradiction. Suppose $f(c) \neq 0$ for all $c \in A$. Let $a_1 = a$ and let $b_1 = b$. Then, we have either $f(\frac{a_1+b_1}{2}) > 0$ or $f(\frac{a_1+b_1}{2}) < 0$.

(1) If $f(\frac{a_1+b_1}{2}) > 0$, set $a_2 = a$ and $b_2 = \frac{a_1+b_1}{2}$.
(2) If $f(\frac{a_1+b_1}{2}) < 0$, set $a_2 = \frac{a_1+b_1}{2}$ and $b_2 = b$.

In either case, we have $f(a_2) < 0 < f(b_2)$.

Similarly, we can find sequences (a_n) and (b_n) such that for all $n \in \mathbb{N}$,

$$f(a_n) < 0 < f(b_n)$$

with the following properties:

(1) $a_n \leq a_{n+1}$ and $b_{n+1} \geq b_n$;
(2) $b_n - a_n = \frac{1}{2^{n-1}}(b_1 - a_1)$.

Step 2: Show the limits (a_n) and (b_n) exist

The first condition and Monotone Convergence Theorem implies that $\lim\ a_n$ exists, and similarly, the second condition and Monotone Convergence Theorem implies that $\lim\ b_n$ exists. The second condition implies that $\lim\ (b_n - a_n) = 0$ and so $\lim\ b_n = \lim\ a_n$.

Step 3: Find limits of $f(a_n)$ and $f(b_n)$ Let $c = \lim\ a_n = \lim\ b_n$. Since f is continuous, $\lim\ f(a_n) = \lim\ f(b_n) = f(c)$. On the other hand, $f(a_n) < 0$ for all n and so $\lim_{n\to\infty} f(a_n) \leq 0$, and $f(b_n) \geq 0$ for all n, and so $\lim_{n\to\infty} f(b_n) = 0$ for all n. Hence, we must have $f(c) = 0$, and so have a contradiction. □

Theorem 5.7. *(Bolzano's intermediate value theorem) Let I be an interval and let $f : I \to \mathbb{R}$ be continuous. Let $a, b \in I$ with $f(a) < f(b)$, and let $k \in (f(a), f(b))$. Then, there exists $c \in I$ between a and b such that $f(c) = k$.*

Proof. Let $g(x) = f(x) - k$. Then $g(a) < 0$ and $g(b) > 0$. Since g is still continuous, by Theorem 5.6, there exists c such that $g(c) = 0$. Hence, $f(c) = k$. □

Example 5.8. Let $f(x) = x^6 - 10x + 1$. Show that f has at least two real roots.
Sol: Note that $f(100) > 0$ and $f(-100) > 0$. We also have $f(1) = -8$. By the Bolzano intermediate theorem, there is $c_1 \in (1, 100)$ such that $f(c_1) = 0$ and there is $c_2 \in (-100, 1)$ such that $f(c_2) = 0$.

Exercise 5.9. Use Bolzano's intermediate value theorem to show that every polynomial of odd degree with real coefficients has at least one root.

Exercise 5.10. Let $f(x) = (x-1)(x-2)(x-3)(x-4)(x-5)$. Use bisection method on the interval $[0, 7]$. Which roots will be located by the bisection method?

5.3. **Continuous functions preserving intervals.**

Corollary 5.11. *Let I be an interval. Let $f : I \to \mathbb{R}$ be a continuous function. Then, $f(I)$ is also an interval.*

Proof. Let $\alpha, \beta \in f(I)$ with $\alpha < \beta$. Then, $f(a) = \alpha$ and $f(b) = \beta$ for some $a, b \in I$. Now, for any $\gamma \in [\alpha, \beta]$, Bolzano's intermediate value theorem implies that $f(c) = \gamma$ for some $c \in I$. This implies that $[\alpha, \beta] \subset f(I)$. This implies that $f(I)$ is an interval $\square$

Example 5.12. Let $f(x) = \sin(x)$. Then $f(\mathbb{R}) = [-1, 1]$, $f([0, \pi]) = [0, 1]$, $f((0, \pi)) = (0, 1]$.

Exercise 5.13. Let I be an interval. Let $f : I \to \mathbb{R}$ be a contnuous function. Determine if the following statements are true:

(1) Let I be a closed bounded interval. Then $f(I)$ is also a closed bounded interval.
(2) Let I be an open bounded interval. Then $f(I)$ is an open bounded interval.

6. Uniform continuity

6.1. **Definitions.**

Definition 6.1. Let $f : A \to \mathbb{R}$. We say that f is uniformly continuous if for any given $\epsilon > 0$, there exists $\delta > 0$ such that for any $x, y \in A$, if $|x - y| < \delta$,

$$|f(x) - f(y)| < \epsilon.$$

Exercise 6.2. Show that a uniformly continuous function $f : A \to \mathbb{R}$ is continuous on A.

Exercise 6.3. Compare with the definition of a continuous function. Do you know why the term 'uniform' is used?

Example 6.4.
(1) Define $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = x$. Then f is uniformly continuous. To see this, for any given $\epsilon > 0$, we choose $\delta = \epsilon$ such that for any $|x - y| < \delta = \epsilon$, $|f(x) - f(y)| = |x - y| < \epsilon$.
(2) Define $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = x^2$. Then f is not uniformly continuous. We choose $\epsilon = 1$. For any $\delta > 0$, we choose $x = \frac{1}{\delta}$ and $y = \frac{1}{\delta} + \frac{\delta}{2}$. Then $|x - y| < \delta$. But $|f(x) - f(y)| = |(x - y)(x + y)| = \frac{\delta}{2}(\frac{2}{\delta} + \frac{\delta}{2}) > 1$.
(3) Define
$$f(x) = \begin{cases} 1 & x > 0 \\ -1 & x \leq 0 \end{cases}$$
Since f is not continuous at 0, f is not uniformly continuous.

Exercise 6.5. Draw a diagram to understand why example (2) above fails to be uniformly continuous.

Exercise 6.6. Let $f : A \to \mathbb{R}$ and let $g : A \to \mathbb{R}$ be uniformly continuous. Show that $f + g$ is also uniformly continuous.

In general, for two uniformly continuous functions $f, g : A \to \mathbb{R}$, fg is not necessarily uniformly continuous.

6.2. **Non-uniform criteria.** As before, one may formulate some sequential criteria, which is useful to show non-uniform continuity:

Exercise 6.7. (Non-uniform Continuity) Let $f : A \to \mathbb{R}$. Then the following statements are equivalent:

(1) f is not uniformly continuous on A.
(2) There exists $\epsilon_0 > 0$ such that for any $\delta > 0$, there exist $x_\delta, y_\delta \in A$ such that $|x_\delta - y_\delta| < \delta$ and $|f(x_\delta) - f(y_\delta)| \geq \epsilon_0$.
(3) There exists $\epsilon_0 > 0$ such that there exist two sequences (x_n) and (y_n) such that $\lim(x_n - y_n) = 0$, and for all $n \in \mathbb{N}$, $|f(x_n) - f(y_n)| \geq \epsilon_0$.

6.3. **Uniform continuous theorem.** We shall introduce a family of uniformly continous functions:

Theorem 6.8. *(Closed+bounded+continuous $\Rightarrow$ Uniformly continuous) Let I be a closed bounded interval. If $f : I \to \mathbb{R}$ is continuous, then f is uniformly continuous.*

Proof. We shall prove by contradiction. Suppose f is not uniformly continuous.
Step 1: Construct two subsequences
By Exercise 6.7, there exists two sequences (x_n) and (y_n) such that $\lim(x_n - y_n) = 0$ and $|f(x_n) - f(y_n)| \geq \epsilon_0$ for some $\epsilon_0 > 0$.

Step 2: Construct convergent subsequences
By Bolzano-Weiterstrass Theorem , let (x_{n_k}) be a convergent subsequence of (x_n). By the Bolzano-Weiterstrass Theorem, there exists a convergent subsequence of (y_m) of (y_{n_k}). Then (x_m) is a subsequence of (x_{n_k}) and so is also convergent.
Step 3: Use continuity
Since f is continuous, $(f(x_m))$ and $(f(y_m))$ converges to $f(z)$. Thus,

$$\lim_{m\to\infty}(f(x_m) - f(y_m)) = 0$$

However, this contradicts to Step 1, since $f(x_m) - f(y_m) \geq \epsilon_0$ implies $\lim_{k\to\infty}(f(x_m) - f(y_m)) \geq \epsilon_0$. □

Example 6.9. The following functions are uniformly continuous:

(a) $f : [1, 2] \to \mathbb{R}$, $f(x) = x^2$
(b) $f : [0, 100] \to \mathbb{R}$, $f(x) = \frac{\sin(x)}{x^2+1}$
(c) $f : [2, 10] \to \mathbb{R}$, $f(x) = \sin\left(\frac{1}{x}\right)$.

Exercise 6.10. Use the above theorem to conclude that the function $f : [0, \infty) \to \mathbb{R}$ given by $f(x) = \frac{1}{1+x^2}$ is uniformly continuous.

Exercise 6.11. A function $f : \mathbb{R} \to \mathbb{R}$ is said to be periodic if there exists $T \in \mathbb{R}$ such that $f(x + T) = f(x)$ for any $x \in \mathbb{R}$. Show that any continuous periodic function is uniformly continuous.

6.4. **Lipschitz functions.** We now describe another family of uniformly continuous functions.

Definition 6.12. Let $f : A \to \mathbb{R}$ be a function. We say that f is a Lipschitz function if there exists a constant $K > 0$ such that for any $x, y \in A$,

$$|f(x) - f(y)| \leq K|x - y|$$

Exercise 6.13. Understand Definition 6.12 geometrically by interpreting the value $\frac{f(x)-f(y)}{x-y}$ as a slope between points $(x, f(x))$ and $(y, f(y))$.

Example 6.14. Let $f : [1, \infty) \to \mathbb{R}$ given by $f(x) = \sqrt{x}$. Now, for any $x, y \geq 1$,

$$|f(x) - f(y)| = |\sqrt{x} - \sqrt{y}| = \frac{|x - y|}{\sqrt{x} + \sqrt{y}} \leq \frac{1}{2}|x - y|,$$

where the last inequality follows from

$$\frac{1}{\sqrt{x} + \sqrt{y}} \leq \frac{1}{2}.$$

Hence, f is Lipschitz.

Example 6.15. Let $g : [0, 2] \to \mathbb{R}$ given by $g(x) = \sqrt{x}$. For $0 \leq x < y \leq 2$,

$$|g(x) - g(y)| = |\sqrt{x} - \sqrt{y}| = \frac{|x - y|}{\sqrt{x} + \sqrt{y}}.$$

Since x, y can be very small, there is no bound for $\frac{1}{\sqrt{x}+\sqrt{y}}$. Hence, f is not Lipschitz.

Theorem 6.16. *Let $f : A \to \mathbb{R}$ be a Lipschitz function. Then f is uniformly continuous.*

Proof. Since f is a Lipschitz function, there exists a constant $K > 0$ such that for any $x, y \in A$,

$$|f(x) - f(y)| \leq K|x - y|$$

Now, for any given $\epsilon > 0$, we choose $\delta = \frac{\epsilon}{2K}$. Then, for any $x, y \in A$ satisfying $|x - y| < \delta$. Then

$$|f(x) - f(y)| \leq K|x - y| < K\delta = \epsilon.$$

Hence, f is uniformly continuous. □

Example 6.17. Let $f : [0, 2] \to \mathbb{R}$ given by $f(x) = \sqrt{x}$. Then f is not Lipschitz, but is uniformly continuous.

6.5. **Cauchy sequence criteria and Extension Theorem.**

Theorem 6.18. *Let $f : A \to \mathbb{R}$ be a uniformly continuous function. If (x_n) is a Cauchy sequence in A, then $(f(x_n))$ is a Cauchy sequence in $\mathbb{R}$.*

Proof. Suppose (x_n) is a Cauchy sequence in A. Let $\epsilon > 0$.
By the definition of uniform continuity, there exits $\delta > 0$ such that for any $x, y \in \mathbb{R}$ satisfying $|x - y| < \delta$,

$$|f(x) - f(y)| < \epsilon.$$

Since (x_n) is Cauchy, there exists $N \in \mathbb{N}$ such that for any $n, m > N$ such that

$$|x_n - x_m| < \delta.$$

Hence, combining above, for any $n, m > N$,

$$|f(x_n) - f(x_m)| < \epsilon.$$

□

Example 6.19.
(1) Let $f(x) = \sin\left(\frac{1}{x}\right)$ for $x \neq 0$. The function is not uniformly continuous on $(0, b]$ for any $b > 0$.
(2) Let $f(x) = x$ and $g(x) = \sin(x)$. Show that the product fg is not uniformly continuous.

Theorem 6.20. *Let $f : (a, b)$ be a continuous function on an open interval (a, b). A function f is uniformly continuous on (a, b) if and only if $\lim_{x\to a} f$ exists and $\lim_{x\to b} f$ exist.*

Proof. If direction
Since f is defined on $[a, b]$, it is a continuous function on closed bounded interval. Then f is uniformly continuous by Theorem 6.8.

Only if direction
Suppose f is uniformly continuous on (a, b). Let (u_n) be a convergent sequence in (a, b) with $\lim\ u_n = a$. It is shown in the previous chapter that (u_n) also forms a Cauchy sequence. Thus, by Theorem 6.18, $(f(u_n))$ is also a Cauchy sequence. Now, it is shown in the previous chapter that $(f(u_n))$ is also convergent.

Now, for any convergent sequence (x_n) with $\lim\ x_n = a$, since $\lim(x_n - u_n) = 0$, the uniform continuity implies

$$\lim\ (f(x_n) - f(u_n)) = 0$$

Hence, $\lim\ f(x_n) = \lim\ (f(x_n) - f(u_n)) + \lim\ f(u_n) = \lim\ f(u_n)$. Hence, f is continuous at a. □

Exercise 6.21. Reinterpret Example 6.19 by Theorem 6.20.

Exercise 6.22. For $b > 0$, let $g : (0, b) \to \mathbb{R}$ given by $g(x) = x \sin\left(\frac{1}{x}\right)$. Since $\lim_{x\to 0} g = 0$ exists, one can extend g to a continuous function on $[0, b]$. Hence, the function g is uniformly continuous .

6.6. **More examples.**

Exercise 6.23. Determine if the following functions are uniformly continuous. Explain breifly:

(1) $f : [0, \frac{\pi}{4}] \to \mathbb{R}$ given by $f(x) = \tan(x)$;
(2) $g : [\frac{\pi}{4}, \frac{\pi}{2}) \to \mathbb{R}$ given by $g(x) = \tan(x)$;
(3) $h : \mathbb{R} \to \mathbb{R}$ given by $h(x) = \frac{1}{2+\cos(x)}$.